L'Union Fait La Force: Participatory Approaches Key for NLP Equality

ALLEN DUFORT, Brown University, USA

CCS Concepts: • **Computing methodologies** → **Natural language processing**; **Language resources**; **Discourse, dialogue and pragmatics**; *Artificial intelligence*; Machine translation; Lexical semantics.

Additional Key Words and Phrases: Low-resource, natural language processing, participation, participatory approach

1 Introduction

Have you ever looked for resources for a particular language only to come up empty-handed? I have. This hypothetical, based on my personal experiences, highlights a prominent issue: there is currently a large disparity between languages in natural language processing (NLP) resources. This dilemma is discussed by Joshi et al. in their paper "The State and Fate of Linguistic Diversity and Inclusion in the NLP World", where they assess the quality of NLP resources across languages and call on computational linguistic researchers to address this disparity [Joshi et al. 2020]. This issue is especially important to me because there are few NLP resources available for Haitian Creole, my family's native language. There are very few if any, well-established online courses for Haitian Creole speakers to learn English or any other language. I did not even realize the extent of the problem until I tried–and failed–to find an online resource to help a family member who recently moved to the United States learn English!

The goal of my project is to develop methods to improve linguistic diversity in NLP resources. "*L'union fait la force*" is Haiti's national motto, translating to "*unity is strength*" or more colloquially "*together we are strong*." I featured this phrase in the beginning of the title because it represents the argument of the paper: in order to minimize the disparity in NLP resources between languages, we must actively involve low-resource language communities in developing NLP tools and resources, extending the involvement beyond the typical methods of human feedback and data collection. I plan on doing this by analyzing instances of people creating linguistic resources for low-resource languages.

2 Literature Review

2.1 Frameworks

I have found numerous papers with frameworks analyzing inequality in linguistic resources. The first paper, "A Ladder of Participation", asserts that if everyone is to be equal in society, society must be restructured so that marginalized groups have the power to make decisions and voice their concerns about policies that affect them [Arnstein 1969]. Arnstein's paper is considered to

Author's Contact Information: Allen Dufort, allendufort2020@gmail.com, allen_dufort@brown.edu, Brown University, Providence, Rhode Island, USA.

ACM XXXX-XXXX/2026/6-ART
https://doi.org/10.1145/nnnnnnn.nnnnnnn

be the beginning of participatory research, conceptualizing participation in a way that can be applied to a variety of situations, including NLP. Another fundamental paper in contextualizing NLP resources is "The State and Fate of Linguistic Diversity and Inclusion in the NLP World" [Joshi et al. 2020]. This paper proposes that languages differ in the availability of NLP resources and categorizes them accordingly to demonstrate these disparities, building on the concepts of the Ladder of Participation. The core framework that I am using is "The Participatory Turn in AI Design: Theoretical Foundations and the Current State of Practice" [Delgado et al. 2023]. This paper applies Arnstein's concepts of participatory design to AI and ML systems by transforming the Ladder of Participation into a framework called Parameters of Participation. I plan on adapting the Parameters of Participation framework to fit the context of low-resource languages. I also draw inspiration from Suresh et al., who apply the Parameters of Participation framework to AI foundation models [Suresh et al. 2024]. Other sources that shape how I want to conceptualize low-resource languages are Te Hiku Media [Hao 2022] and Masakhane [Masakhane 2024]. These organizations focus on improving NLP resources for Maōri and African languages respectively. The most important aspect of both projects is their emphasis on the language communities' data sovereignty, or the right to decide how their data are collected and used. This principle is directly tied to the Ladder of Participation, which argues that collaboration and ownership are necessary for marginalized groups to achieve equality.

2.2 Case Studies

I have numerous sources that address the need for more linguistic diversity in NLP. Te Hiku Media has a project that creates NLP resources to help revive the Māori language [Hao 2022]. Masakhane has multiple projects focusing on including native speakers of African languages in their research to improve their NLP resources [Nekoto et al. 2020; Wang et al. 2023]. These projects incorporate native speakers at every step of the NLP process, including data collection, data distribution, algorithm development, benchmark development, model evaluation, and more. I also have sources that do not have direct input from the communities of the low-resource languages they are studying [Cahyawijaya et al. 2024; Remy et al. 2024; Yong et al. 2024]. Researchers in these sources instead rely on more traditional methods to improve NLP resources and ML accuracy for low-resource languages, such as fine-tuning.

2.3 Review Conclusion

I included all these articles to gain a more holistic view of current approaches for improving the NLP resources of low-resource languages. I plan to use my references to demonstrate that participatory principles are necessary to enhance low-resource languages in NLP.

3 Methodology

3.1 Parameters of Participation in NLP

For my analysis, I use the Parameters of Participation, a conceptual framework Delgado et al. designed for analyzing participatory approaches to AI design [Delgado et al. 2023]. In this paper, I edit this framework to be more applicable to NLP, calling it "The NLP Participatory Matrix" (see Table 2). The *dimensions of participation* in my framework are the *goal* of the NLP project (why is participation needed?), the *scope* (who is involved?), and the *form* (what form does participation take?), coming directly from the Parameters of Participation [Delgado et al. 2023]. The *modes of participation* describe each dimension of the participatory approaches. Suresh et al. define the *modes of participation* in these four parts: "*consultation* (e.g., eliciting user preferences), *inclusion* (e.g., deliberation around specific design choices), *collaboration* (e.g., co-creation of design possibilities), and *ownership* (e.g., stakeholders shape the entire design process)" [Suresh et al. 2024]. I plan on using the dimensions and modes of participation defined in these two papers, but I add a fifth mode of participation, *no participation*, in my framework for cases where projects have little to no participatory approaches.

3.2 Case Studies Background

This subsection provides relevant details about the NLP projects (e.g. research goals) analyzed in section 3.3.

3.2.1 Tweeties. Tweeties is a research project that aims to improve low-resource NLP applications, such as language modeling and translations, without requiring high amounts of data [Remy et al. 2024]. This project uses trans-tokenization, a cross-lingual vocabulary transfer method, to tailor large language models (LLMs) trained on high-resource languages to low-resource languages. Their results show that the trans-tokenization models perform similarly to other models that have significantly more data, but cultural biases from the high-resource languages are transferred to the LLMs for the low-resource languages.

3.2.2 LexC-Gen. LexC-Gen is a research project run by Brown University [Yong et al. 2024]. This project uses LLMs and bilingual lexicons between high and low-resource languages to generate synthetic data for low-resource languages. LexC-Gen performs better than some open-source LLMs and at similar levels to gold standard translations, but has some technical issues such as mismatch syntax.

3.2.3 Cendol. Cendol is a collection of LLMs tailored to high and low-resource Indonesian languages [Cahyawijaya et al. 2024]. Cendol aims to improve NLP tasks, such as translations, paraphrasing, and human alignment, by using instruction tuning and adapting the LLMs to the local languages. Although Cendol has shown improvement from previous NLP projects for Indonesian languages, the project still has some limitations including a lack of advanced human alignment and capturing cultural nuances localized to a particular language.

3.2.4 AfriCOMET. AfriCOMET is a project that seeks to improve machine translation (MT) resources for low-resource African languages by creating a new framework to collect evaluation data and evaluation metrics tailored to African languages [Wang et al. 2023].

3.2.5 Masakhane. Masakhane is a grassroots organization whose goal is to strengthen and increase NLP research in African languages through participatory approaches, like open access to data and working with people from multiple disciplines [Masakhane 2024]. Masakhane employs a participatory approach to develop MT resources, as outlined in a study by Nekoto et al. [Nekoto et al. 2020]. This approach actively involves native speakers, researchers, and local communities in the MT development process.

3.2.6 Te Hiku. Te Hiku Media is a nonprofit Māori radio station based in New Zealand [Hao 2022]. The goal of their NLP project is to revive the Māori language *te reo* while ensuring that the local Māori communities control their data.

3.3 Case Studies: Analysis

This section analyzes commonalities between the case studies in each category and the defining features of each category. A summary of this analysis is illustrated in Table 2.

3.3.1 No participation. There are some commonalities between the two projects that have no participation [Remy et al. 2024; Yong et al. 2024]. Neither project orients its goals to engage the local communities of the low-resource languages being studied in the research. Instead, both projects try to make resources with limited data, deprioritizing participation from low-resource language communities. As a result, there is no effort to even contact low-resource language communities and those communities have no involvement in the projects.

3.3.2 Consultation. I am putting Cendol in the *consultation* category because it only seeks to gain feedback and linguistic nuances from the local communities of the Indonesian languages they study [Cahyawijaya et al. 2024]. In terms of participatory approaches, Cendol works with both NLP experts and non-experts but does not go beyond gathering input from participants.

3.3.3 Inclusion. AfriCOMET's objective is to improve MT models for African languages by modifying evaluation metrics to better fit African languages [Wang et al. 2023]. They include both NLP experts and non-experts native speakers of African languages in the project in the collection and evaluation of the data. However, I am placing AfriCOMET in the *inclusion* category versus *collaboration* or *ownership* because the participants do not have control over the design of the project, such as the evaluation metrics.

3.3.4 Collaboration. Masakhane's goal in Nekoto et al.'s study is to improve MT for low-resource languages using participatory approaches [Nekoto et al. 2020]. This project in particular is in the realm of *collaboration* because it aims to work together with a diverse group of people from local communities to achieve their goal. People from the local communities not only collect, translate, and evaluate the data, but they also share ownership of the data and models with Masakhane. I consider this more *collaboration* than *ownership* since the local communities do not have full ownership of the project, but share it with Masakhane.

3.3.5 Ownership. The purpose of Te Hiku's project is to revive *te reo*. Similar to Masakhane, Te Hiku recruits a wide variety of people to achieve its goal. I am designating this project as *ownership* instead of *collaboration* because the Māori people have the final say in decisions regarding the project's direction and data access.

4 Discussion

The projects with high levels of involvement (i.e. those under *collaboration* and *ownership*) tend to perform better than those with lower participation. Projects with low participation tend to have technical issues and limitations. Primarily, when the projects have limited participation, the NLP models carry over cultural biases from high-resource languages [Cahyawijaya et al. 2024; Remy et al. 2024]. Secondly, they tend to have mixed results; for instance, some languages show no improvement in MT models using AfriCOMET compared to using other evaluation metrics [Wang et al. 2023]. The high-participation projects seem to not run into these issues as much. I theorize that the main reason for this is because: 1. having native speakers create and evaluate the data removes the biases from high-resource languages and 2. involving people from numerous disciplines allows for new perspectives and ideas. Overall, the higher the *mode of participation* is, the more robust the project is.

5 Recommendations

Based on my findings, I recommend incorporating *collaboration* or *ownership* in your project if you are researching low-resource languages. If you do not fully incorporate the communities of the languages you are studying, you run the risk of not including the appropriate linguistic and cultural aspects in your NLP application. Language is so much more than just words; it is created and sustained by communities, which must be accounted for.

No participation	Consultation	Inclusion	Collaboration	Ownership

Table 1. Legend for Table 2

Project Name	**Goal**	**Scope**	**Form**
Tweeties	Adapt LLMs for high-resource languages to low-resource languages	No involvement	No involvement
LexC-Gen	Generate synthetic labeled data for low-resource languages	No involvement	No involvement
Cendol	Align LLMs with human preferences	Native speakers of Indonesian languages	Giving feedback to researchers
AfriCOMET	Improve MT resources for African languages	Native speakers of African languages	Collecting and evaluating MT data
Masakhane	Improve MT for low-resource African languages	Native speakers of African languages	Translating, curating data, evaluating models
Te Hiku	Revive the *te reo* language	Māori community	Determining project goals, designing project, controlling data access

Table 2. NLP Participatory Matrix for case studies

There is an endless number of ways to incorporate *collaboration* or *ownership* in NLP projects but I will go over the following points of incorporation in my recommendations: goal setting, researcher selection, data collection, model development, evaluation, project deployment, and data access. The execution of *collaboration* and *ownership* in an NLP project is very similar, but the main difference is that with *ownership*, low-language communities have more decision-making power than the initial researchers. If *ownership* is not feasible in your project, then I recommend employing *collaboration* as your project's *mode of participation.*

5.0.1 Goal setting. To have *collaboration* or *ownership*, members of the low-resource language communities should participate in determining the project's goals, the desired outcomes, the desired community impact, and more. To have the project at least be collaborative, the low-resource language communities' opinions and desires should have equal consideration to the other researchers in the project.

5.0.2 Researcher selection. Low-resource language communities should have a central role in selecting the other researchers on the project. The low-resource language communities may include people with local cultural nuances and different perspectives from typical researchers.

5.0.3 Data collection. To have *collaboration* or *ownership* in data collection, the low-resource language communities of the research project should determine what data is getting collected, how it is collected, who collects the data, and the reasoning behind data collection.

5.0.4 Model development. For model development, I recommend that the low-resource language communities determine what AI models are used, if any. They should also be in charge of the actual execution of the model development, including model training, testing, and evaluation.

5.0.5 Evaluation. I recommend that for evaluation, the low-resource language communities design the evaluation metrics, how the quality of the results will be determined, and who will evaluate the data. Evaluation is particularly important as linguistic projects can have subjective outputs that should be evaluated by people from the languages' local communities.

5.0.6 Project deployment. In participatory terms of *collaboration* and *ownership*, the low-resource language communities should be in charge of how the project is deployed, like what companies can use the project, which communities will be impacted or targeted by the project, monetary policies, and feedback collection from stakeholders.

5.0.7 Data access. Ownership and *collaboration* over data access would look like having shared ownership over the datasets, models, and results. This also includes controlling who has access to the data.

Acknowledgments

Thank you to Professor Harini Suresh, for her advice and good sources. Thanks also to the SOLAR lab in Brown University's Computer Science Department for their good sources on NLP.

References

Sherry R. Arnstein. 1969. "A Ladder Of Citizen Participation." *Journal of the American Institute of Planners*, 35, 4, 216–224. eprint: https://doi.org/10.1080/01944366908977225. DOI: 10.1080/01944366908977225.

Samuel Cahyawijaya et al.. 2024. "Cendol: Open Instruction-tuned Generative Large Language Models for Indonesian Languages." *arXiv preprint arXiv:2404.06138*. https://arxiv.org/abs/2404.06138.

Fernando Delgado, Stephen Yang, Michael Madaio, and Qian Yang. 2023. "The Participatory Turn in AI Design: Theoretical Foundations and the Current State of Practice." In: *Proceedings of the 3rd ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization* (EAAMO '23) Article 37. Association for Computing Machinery, Boston, MA, USA, 23 pages. ISBN: 9798400703812. DOI: 10.1145/3617694.3623261.

Karen Hao. 2022. "A new vision of artificial intelligence for the people." *MIT Technology Review*. https://www.technologyreview.com/2022/04/22/1050394/artificial-intelligence-for-the-people/.

Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. July 2020. "The State and Fate of Linguistic Diversity and Inclusion in the NLP World." In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Ed. by Dan Jurafsky, Joyce Chai, Natalie Schluter, and Joel Tetreault. Association for Computational Linguistics, Online, (July 2020), 6282–6293. DOI: 10.18653/v1/2020.acl-main.560.

Masakhane. 2024. *About Masakhane*. Accessed: 2024-12-01. (2024). https://www.masakhane.io.

Wilhelmina Nekoto et al.. Nov. 2020. "Participatory Research for Low-resourced Machine Translation: A Case Study in African Languages." In: *Findings of the Association for Computational Linguistics: EMNLP 2020*. Ed. by Trevor Cohn, Yulan He, and Yang Liu. Association for Computational Linguistics, Online, (Nov. 2020), 2144–2160. DOI: 10.18653/v1/2020.findings-emnlp.195.

François Remy, Pieter Delobelle, Hayastan Avetisyan, Alfiya Khabibullina, Miryam de Lhoneux, and Thomas Demeester. 2024. "Trans-tokenization and cross-lingual vocabulary transfers: Language adaptation of LLMs for low-resource NLP." *arXiv preprint arXiv:2408.04303*. https://arxiv.org/abs/2408.04303.

Harini Suresh, Emily Tseng, Meg Young, Mary Gray, Emma Pierson, and Karen Levy. 2024. "Participation in the age of foundation models." In: *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency* (FAccT '24). Association for Computing Machinery, Rio de Janeiro, Brazil, 1609–1621. ISBN: 9798400704505. DOI: 10.1145/3630106.3658992.

Jiayi Wang et al.. 2023. “AfriMTE and AfriCOMET: Empowering COMET to Embrace Under-resourced African Languages.” *arXiv preprint arXiv:2311.09828*. https://arxiv.org/abs/2311.09828.

Zheng-Xin Yong, Cristina Menghini, and Stephen H Bach. 2024. “LexC-Gen: Generating Data for Extremely Low-Resource Languages with Large Language Models and Bilingual Lexicons.” *arXiv preprint arXiv:2402.14086*. https://arxiv.org/abs/2402.14086.